

Fig. 2: Pin configuration of alphanumeric display

at a time, respectively. However, displays DIS1 and DIS2 are enabled or disabled directly by port pins P1.0 and P1.1. Pins 4 and 5 are grounded and pin 6 is made high to enable decoder 74LS138.

Fig. 2 shows the pin configuration of the common-anode alphanumeric display.

All the corresponding data pins DIS1 through DIS16 of alphanumeric displays have been tied together, while the common anode of each display is separately powered via a BC558 transistor which switches 'on' or 'off' as required, through outputs of 74LS138 ICs and pins P1.0 and P1.1 of IC1. The higher nibble of port P3 (P3.4 through P3.7) is used as a selection bus to select one of the 16 previously stored messages using the 4-bit binary value present on these pins. This value can be changed through a 4-pin DIP switch (S0 through S3).

Selection pins P3.4 through P3.7 are pulled high via resistors R36 through R33, respectively. When the switch connected to a given pin is open the value is high (1), and when it is closed the pin is held low and the value becomes '0.' In this way, by using a 4-bit number you can select any of the 16 messages shown in the Table.

Capacitor C5 and resistor R37 form the power-'on' reset circuit, while a push-to-connect switch has been used for manual reset. An 11.0592MHz crystal generates the basic clock frequency for the microcontroller. To change the message being displayed while the circuit is working, first change the number present at the selection bus, then press 'reset' key.

The 220V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 500 mA. The output of the transformer is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C3 and then regulated by IC 7805 (IC4). Capacitor C4 bypasses any ripple present in the regulated power supply. LED1 acts as the power-'on' indicator.

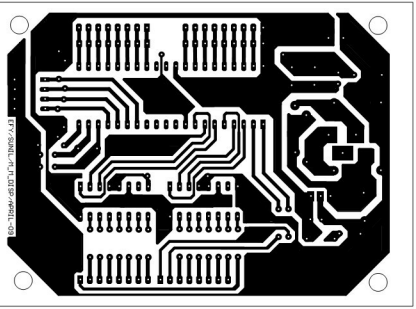


Fig. 3: Actual-size, single-side PCB for the microcontroller-based moving-message display circuit, except alphanumeric display and associated components

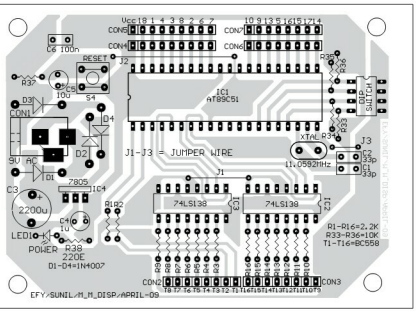


Fig. 4: Component layout for the PCB shown in Fig. 3

Construction

Fig. 3 shows an actual-size, single-side PCB layout for the microcontroller-based moving-message display circuit, except displays DIS1 through DIS16, transistors T1 through T16 and resistors R17 through R32. Com-